

SUPPLEMENTAL MATERIAL TO
Failures of Contingent Thinking

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In this supplemental material to [Piermont and Zuazo-Garin \(2026\)](#), we explore provide additional examples of how our theory relates to recent experimental findings. **Framing effects.** Framing effects, wherein different but logically equivalent descriptions of a choice problem yield different choices, constitute a simple and evident violation of rational decision-making. For example, consider a DM valuing bets over the following statements:

P = “the S&P500 goes up tomorrow”, or

Q = “the S&P500 goes up tomorrow and the 1 millionth prime number is greater than 15,000,000.”

While P and Q are in fact equivalent, as the additional criterion in Q is tautological, it is far from unreasonable for a DM to strictly prefer betting on the former. As humans are not perfect reasoners, we often fail to observe logical equivalence, and thus do not treat logically equivalent statements as such ([Stalnaker, 1991](#); [Lipman, 1999](#)). This is easily captured in our model by a t that maps P and Q to.

While the effect above is likely due to the computational complexity of the statements, framing effects abound for myriad other, often emotional, reasons. For example, [Tversky and Kahneman \(1981\)](#) asked subjects consider the efficacy of a drug differently depending on its description.

- 100 patients took the medicine, and 68 patients found it beneficial, or
- 100 patients took the medicine, and 32 patients saw no improvement.

Again, such differentiation must stem from the failures of of the DM to identify different descriptions of the same event.

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Redundant Evidence. In a recent working paper, [Garfagnini and Walker-Jones \(2023\)](#) endow subjects with a lottery that has a $x \in \{0, 1, 2, \dots, 100\}$ percent chance of paying \$20 and a $1 - x$ percent chance of paying nothing. Subjects do not know x but know that each value of x is equally likely. The experimenters elicit from the subjects their *probability-equivalent*, that is, the minimum $y \in [0, 100]$ such that they are willing to trade the unknown (or compound) lottery for a y percent chance of \$20. This equivalent is elicited before and after providing the subjects with various pieces of information about the initial, endowed, lottery of the form

$$P_n = "x \text{ is greater than } n"$$

Within this rather simple environment, subjects displayed clear departures from rational behavior. In particular, they often changed their valuation upon receiving redundant evidence. Specifically, they are initially told P_n , reveal a probability-equivalent y , and are subsequently told $P_{n'}$, with $n' \leq n$, and reveal a *new* probability-equivalent $y' \neq y$.

Here, clearly, the subjects' behavior is inconsistent with an understanding that P_n implies $P_{n'}$. After conditioning on P_n , the subjects should consider only states where $x \geq n$ so that the additional revelation of $P_{n'}$ provided no new information. There are two (not necessarily mutually exclusive) explanations: the first is that in the initial state-space, there are inconsistent states, where P_n is true, but $P_{n'}$ is false. The second explanation is that subjects do not update by excluding states, but use some other rationally-limited heuristic.

A further result from the authors suggests strongly that the latter is a contributing factor: subjects often revise their probability-equivalents in the opposite direction as expected. Over 60% of subjects with a baseline probability-equivalent y *lowered* their response to $y' < y$ after being informed of P_{20} (or erred analogously in the opposite direction). While properly disentangling static vs. dynamic rationality would require a more intentional task, this evidence indicates that reducing uncertainty is not necessarily accompanied by a higher standard of rational thinking.

What you see is all there is. The following is a slight adaption of [Enke \(2020\)](#): Subjects are presented with bets regarding 6 independent, uniform draws from $X = \{-3, -2, -1, 1, 2, 3\}$. Subjects observe the first draw and subsequently indicate whether

they believe the average to be positive or negative. Thereafter, they observe additional signals by interacting with a computerized information source that (transparently) shares all signals that “align” with their first stated belief (e.g., are positive if the first belief is positive) but not all signals that “contradict” said belief. Subjects then must again state their beliefs about the average. Whenever subjects’ first signal is positive, their final stated beliefs tend to be upward biased, as if they were ignoring the information contained in *not observing* a given signal (and conversely for initially negative assessments).

Consider $X^+ = \{1, 2, 3\}$ and $X^- = \{-1, -2, -3\}$, and without loss of generality assume that the initial draw is $x \in X^+$. Further, assume that a participant guesses that the average is above positive.

Consider the following statements

$\text{OBS}_n = “n \text{ signals are observed}”$

$\text{NEG}_{5-n} = “5 - n \text{ signals are negative}”$

Then it seems that the participant’s behavior is being driven by a failure to perceive the equivalence “ OBS_n if and only if NEG_{5-n} ” for each $n \leq 5$.

To see how this failure of contingent thinking can be captured by an IOU, consider the following state-space: $W = \{\pi : X \rightarrow \mathbb{N} \mid \sum_X \pi(x) = 5\}$, where each state is identified with a function π such that $\pi(x)$ counts the number of draws of x . Thus, t maps the statement, for $x \in X$ and $m \leq 5$,

$\text{D}_{x,m} = “\text{there were } m \text{ draws of } x”$

to the event $\{\pi \in W \mid \pi(x) = m\}$. Now consider the statement, for $x \in X^+$ and $m \leq 5$,

$\text{O}_{x,m} = “\text{there were } m \text{ observations of } x”$

For a perfectly rational subject this should map to the event $\{\pi \in W \mid \pi(x) = m\}$; however, a participant who does not intuit that the missing signals must have been negative will understand this statement as the event $t : \text{O}_{x,m} \mapsto \{\pi \in W \mid \pi(x) \geq m\}$.

So long as the participants are otherwise rational, they will understand the statement OBS_n as mapping to the union of all different ways they could have observed n signals, and likewise the statement NEG_{5-n} as the union of the different ways of

having drawn $5 - n$ negative draws. That is:

$$t(\text{OBS}_n) = \bigcup \left\{ \bigcap_{x \in X^+} t(o_{x,m_x}) \mid \sum_{x \in X^+} m_x = n \right\} \quad \text{and}$$

$$t(\text{NEG}_{5-n}) = \bigcup \left\{ \bigcap_{x \in X^-} t(d_{x,m_x}) \mid \sum_{x \in X^-} m_x = 5 - n \right\}$$

It is straightforward to see that for the rational subjects, these two sets coincide, whereas for subjects with the flawed t given above, $t(\text{NEG}_{5-n}) \subsetneq t(\text{OBS}_n)$.

Support Theory. Tversky and Koehler (1994) collect many examples of *unpacking* wherein DMs estimate the probability of an implicit disjunction (e.g., death from a natural cause) to be less than the sum of its mutually exclusive components (e.g., death from heart disease, cancer, or other natural causes). In other words, subjects assessed the collective probability of a set of specific statements as more likely than a statement comprising their disjunction.

This can be captured by a subject who correctly understands that each component statement is contained in the more general case, but fails to properly account for the mutual exclusivity of statements. Consider the following:

H = “The primary cause of death is heart disease.”

C = “The primary cause of death is cancer.”

$H \wedge C$ = “The primary cause of death is heart disease and the primary cause of death is cancer”

N = “The primary cause of death was natural.”

and assume that the subject understands that $H \wedge C$ is impossible, so that $t(H \wedge C) = \emptyset$, and that both H and C imply N: $t(H) \subseteq t(N)$ and $t(C) \subseteq t(N)$.

Now, if t distributes over conjunctions, it follows that $t(H) \cap t(C) = \emptyset$, and so it must be that $\ell(t(H)) + \ell(t(C)) \leq \ell(t(N))$. However, consider (W, t) with $W = \{w_1, w_2, w_3\}$ and $t : H \mapsto \{w_1, w_2\}, C \mapsto \{w_2, w_3\}, H \wedge C \mapsto \emptyset$ and $N \mapsto W$.

Here, t is not $\wedge$ -distributive, so that although the subject directly recognizes that $H \wedge C$, she still implicitly considers states where neither statement is ruled out.

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